Measuring Hawkish and Dovish Signals in FOMC Communications

A Text-as-Data Analysis of Statements and Minutes, 2000–2024

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Abstract

Central bank communication is an instrument of monetary policy because it shapes expectations beyond the current rate decision. This paper measures the hawkish/dovish signal in Federal Open Market Committee (FOMC) statements and minutes from 2000 to 2024. The main model is an interpretable monetary-policy dictionary that contrasts language associated with inflation pressure, restriction and tightening with language associated with easing, downside risk and labor-market weakness. The lexical measure is complemented by decision-masked scores, sentence-level tagging, TF-IDF classification benchmarks, dictionary sensitivity checks, Latent Semantic Analysis (LSA), and a Sentence-BERT prototype check. The resulting signal is strongly aligned with rate hikes, becomes less hawkish in high-volatility periods, and shows little descriptive difference around election windows. Minutes are not a mechanical repetition of statements: average statement-minute differences are modest, but several meetings display large document-level gaps. The embedding checks show that dense semantic representations capture broader document structure and monetary episodes, while the dictionary remains the most directly interpretable measure of the policy signal.

1 Introduction

The Federal Reserve does not only move the federal funds rate. It also explains economic conditions, signals risks, and manages expectations about the future path of policy. This communication channel is especially visible after FOMC meetings, when the Committee publishes a short statement, followed several weeks later by more detailed minutes. These documents are official, recurrent, and closely read by financial markets. They are therefore a natural corpus for studying monetary policy communication with natural language processing.

This paper asks a text-as-data question: to what extent do FOMC communications send a hawkish or dovish signal, and how does this signal vary across document types, rate decisions, market volatility, election periods, and macroeconomic regimes? A hawkish signal emphasizes inflation, price pressure, restrictive conditions, tightening, or higher rates. A dovish signal emphasizes support to activity, labor-market weakness, easing, lower rates, or downside risks.

The object of interest is latent. It is not directly observed in the data; it must be inferred from words, phrases and document structure. The paper therefore combines an interpretable dictionary model with robustness checks based on standard NLP representations. The dictionary score is the core measure because it maps transparently to monetary-policy concepts. TF-IDF classifiers test whether rate-decision information is also recoverable from a more flexible bag-of-words representation. LSA and Sentence-BERT embeddings test whether dense semantic geometry produces a similar hawkish/dovish ordering or instead captures broader institutional and macroeconomic structure.

The results are descriptive rather than causal. They show that hike meetings are substantially more hawkish than hold or cut meetings, and that high VIX meetings are associated with a less hawkish tone. Election windows do not display a strong average difference. Statements and minutes have similar average tone, but meeting-level distances can be large, which suggests that the minutes often contextualize or qualify the immediate post-meeting message. Finally, combining

Table 1: Aligned corpus description.

Document type	Meetings	Mean words	Median words	Mean sentences	Date range
Statement	199	526.1	536.0	26.8	2000–2024
Minutes	199	7282.7	7518.0	311.9	2000–2024

the hawkish/dovish score with an inflation/employment balance highlights that diagnosis and policy signal are related, especially in minutes, but are not the same textual object.

The contribution is threefold. First, the paper builds a reproducible pipeline that aligns FOMC statements and minutes by meeting and enriches them with financial, political and macroeconomic context. Second, it proposes a transparent measure of monetary-policy tone that can be inspected at the document and sentence levels. Third, it compares this handcrafted measure with more flexible text representations. This comparison is useful because it separates two tasks that are often conflated: measuring an economically defined concept and predicting a policy-decision label from text.

2 Related Work

Central bank communication has become a central component of monetary policy. Blinder et al. [2008] review the theoretical and empirical evidence and argue that communication can make policy more predictable, affect asset prices and shape expectations. This motivates treating FOMC texts as economic data rather than as purely institutional narrative.

Several papers use computational methods to study FOMC communication. Lucca and Trebbi [2009] build automated measures of FOMC statement content and show that long-term interest rates react to communication surprises. Their result is important for the present study because the same rate decision can be accompanied by different language about the future stance of policy. Hansen et al. [2018] apply computational linguistics to FOMC transcripts and show that transparency changes deliberation and communication patterns. This paper focuses instead on the public and regularly available statements and minutes, but it follows the same idea that monetary communication can be measured systematically.

The paper also follows the broader text-as-data literature. Gentzkow et al. [2019] emphasize that textual analysis requires explicit choices about representation, measurement and validation. Loughran and McDonald [2011] show that generic sentiment dictionaries can be misleading in financial corpora, where common words may have domain-specific meaning. This is why the main measure here is not a positive/negative sentiment index, but a monetary-policy tone index. The embedding checks draw on Latent Semantic Analysis [Deerwester et al., 1990] and Sentence-BERT [Reimers and Gurevych, 2019], which provide alternative semantic representations of the same documents.

This paper is closest to dictionary-based work in spirit, but it uses machine learning representations as diagnostics rather than as the main estimator. That choice reflects a practical constraint of the corpus. The sample is small by modern NLP standards: after alignment, the unit of analysis is fewer than two hundred meetings. Without expert sentence labels, a fully supervised hawkish/dovish classifier would risk learning period-specific language or the format of the documents instead of the underlying policy signal. A dictionary is less flexible, but it provides a clear link between the measured score and the economic vocabulary used to construct it.

3 Data

3.1 FOMC Corpus

The corpus contains FOMC statements and minutes collected from the Federal Reserve website [Board of Governors of the Federal Reserve System, 2024]. Statements are the short announcements released immediately after each meeting. Minutes are longer documents published later; they summarize the discussion, the uncertainty around the outlook, and the arguments behind the decision.

The final aligned sample covers 199 meetings from February 2000 to December 2024. Each observation contains a statement, the corresponding minutes, the meeting date and contextual variables. Rate decisions are inferred from the statement text and classified as *hike*, *hold* or *cut*. The sample contains 59 hikes, 110 holds and 30 cuts.

Table 1 shows the large difference in document length. Minutes are roughly fourteen times longer than statements

Table 2: Rate-decision distribution in the aligned sample.

Decision class	Meetings	Share
Cut	30	15.1%
Hold	110	55.3%
Hike	59	29.6%

on average. This makes length normalization essential and also explains why statements and minutes should not be expected to carry the same kind of signal. Statements are calibrated policy messages; minutes are richer institutional summaries.

3.2 Alignment and Preprocessing

The raw corpus is organized around FOMC meeting dates. Statements and minutes are kept as separate fields and then aligned by meeting date. The preprocessing is deliberately conservative. Text is lowercased for counting, whitespace is normalized, and recurring web-page fragments such as navigation text, update footers and accessibility notices are stripped when they appear. The analysis does not remove economically meaningful stopwords from the dictionary scores, because several multi-word expressions depend on local syntax, for example *higher rates*, *downside risks* and *price stability*. Stopword removal is used only in the TF-IDF benchmark, where the goal is predictive comparison rather than direct term accounting.

Documents are also segmented into sentences. Sentence segmentation serves two purposes. It makes it possible to mask sentences that explicitly announce the rate decision, and it allows a sentence-level audit of the dictionary. This step is important because a document-level count can hide the fact that a few repeated words drive the score. Sentence tags make the measure easier to interpret: a hawkish document should contain identifiable hawkish sentences, not only isolated tokens.

Table 2 reports the class balance. The dominance of hold meetings is expected in monetary policy data, but it matters for evaluation: accuracy alone can be misleading. This is why the predictive experiments also report macro-F1, which gives equal weight to the three decision classes.

3.3 Contextual Variables

The empirical analysis adds three sets of contextual variables. First, daily VIX data are retrieved from Cboe [Cboe Global Markets, 2026]. For each FOMC meeting, the VIX is averaged over a window from five days before to five days after the meeting; the upper quartile defines high-volatility meetings. Second, election windows are defined as meetings within 90 days of U.S. presidential or midterm elections. Third, meetings are assigned to broad macro-financial regimes: the early-2000s easing cycle, the global financial crisis, the zero lower bound period, the 2015–2019 normalization, the Covid crisis, the post-Covid inflation period, and a baseline category for the remaining meetings.

These contextual variables are not used to identify causal effects. They are used to organize the descriptive results and to check whether the textual signal moves in economically plausible ways. For example, a useful tone measure should be more hawkish around hikes than around cuts, but it should also reveal more nuanced patterns in stress periods, when the Fed can discuss inflation risks while simultaneously emphasizing financial fragility or downside risks.

4 Methodology

4.1 Dictionary Tone Score

The main model is an expert-guided dictionary score. Let $H_{d,t}$ and $D_{d,t}$ denote the length-normalized counts of hawkish and dovish terms in document d for meeting t:

$$H_{d,t} = \frac{\#\text{hawkish terms}_{d,t}}{\#\text{words}_{d,t}}, \qquad D_{d,t} = \frac{\#\text{dovish terms}_{d,t}}{\#\text{words}_{d,t}}. \tag{1}$$

The normalized tone score is

$$\text{ToneNorm}_{d,t} = \frac{H_{d,t} - D_{d,t}}{H_{d,t} + D_{d,t}}. \tag{2}$$

Positive values indicate a more hawkish document, negative values indicate a more dovish document, and values close to zero indicate a balanced document.

The hawkish dictionary includes terms such as *inflation*, *price stability*, *elevated*, *tightening*, *restrictive*, *higher rates*, *persistent inflation*, and *inflationary pressures*. The dovish dictionary includes terms such as *unemployment*, *weakness*, *slowdown*, *accommodative*, *support*, *lower rates*, *easing*, *recovery*, and *downside risks*. The score is intentionally interpretable: its changes can be audited by reading the terms and the sentences that drive them.

The normalization in Equation (2) is chosen because it is scale-free. A raw difference $H_{d,t} - D_{d,t}$ would still depend on the absolute density of dictionary terms. The normalized balance instead asks whether policy-related language is more hawkish or more dovish conditional on the document containing such language. This is particularly useful when comparing statements and minutes, because minutes contain far more words and far more policy-relevant sentences. When both $H_{d,t}$ and $D_{d,t}$ are zero, the score is set to zero; in practice this is rare for FOMC texts.

4.2 Choice of Model Family

The dictionary score is not meant to be the most predictive possible model. It is chosen because the research question is measurement-oriented. The paper aims to construct an interpretable series that can be compared across meetings and document types. For this goal, a transparent score is preferable to a black-box classifier trained on a small sample.

The additional NLP models answer different questions. TF-IDF with linear models tests whether the text contains enough information to recover the observed rate decision. LSA tests whether documents that are semantically close in a reduced bag-of-words space also look similar on the policy-tone axis. Sentence-BERT tests whether contextual embeddings, without task-specific fine-tuning, align with the same hawkish/dovish prototype contrast. The models are therefore complementary: the dictionary measures an economic concept, while TF-IDF and embeddings evaluate how stable that concept is under alternative representations of the corpus.

4.3 Validation and Robustness Checks

Four checks are used to assess whether the dictionary captures a meaningful policy signal.

First, a masked score removes sentences that directly announce the rate decision. This tests whether the tone is only a restatement of “the Committee raised the target range” or whether it also appears in the surrounding diagnosis and justification.

Second, a sentence-level version tags each sentence as hawkish, dovish, mixed or neutral according to dictionary hits. This provides a qualitative audit and a document-level share of policy-relevant sentences.

Third, TF-IDF benchmarks classify the rate decision from the text. Logistic regression and linear SVM models are trained on chronological splits, using earlier meetings for training and later meetings for testing. This split is stricter than a random split because monetary-policy language changes over time. Models are run on full, decision-masked and boilerplate-stripped text for statements and minutes. The TF-IDF representation uses unigrams and bigrams, English stopword removal, a minimum document frequency of two, and a maximum vocabulary of 6,000 features. The test set contains the last 30% of meetings, covering June 2017 to December 2024; the training set covers February 2000 to May 2017.

Fourth, dense representations are used as semantic robustness checks. LSA is computed by applying truncated SVD to a TF-IDF matrix of document excerpts. A prototype tone is then defined as cosine similarity to a hawkish prototype minus cosine similarity to a dovish prototype. Sentence-BERT uses the same prototype logic with contextual embeddings from `all-MiniLM-L6-v2`. These embedding scores are not treated as replacements for the dictionary; they test whether a less hand-engineered semantic space produces a related ordering.

Finally, the paper uses a simple pooled OLS model as a descriptive check. The dependent variable is the tone score for each document. The explanatory variables are document type, rate-decision dummies, standardized VIX and an election-window indicator. The model is not interpreted structurally. It is a compact way to see whether the associations visible in the figures remain present when these dimensions are included together.

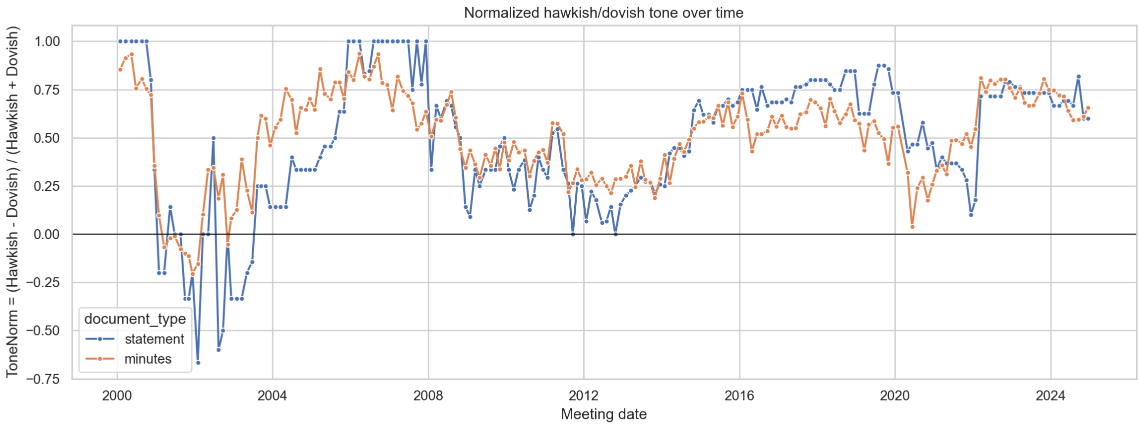
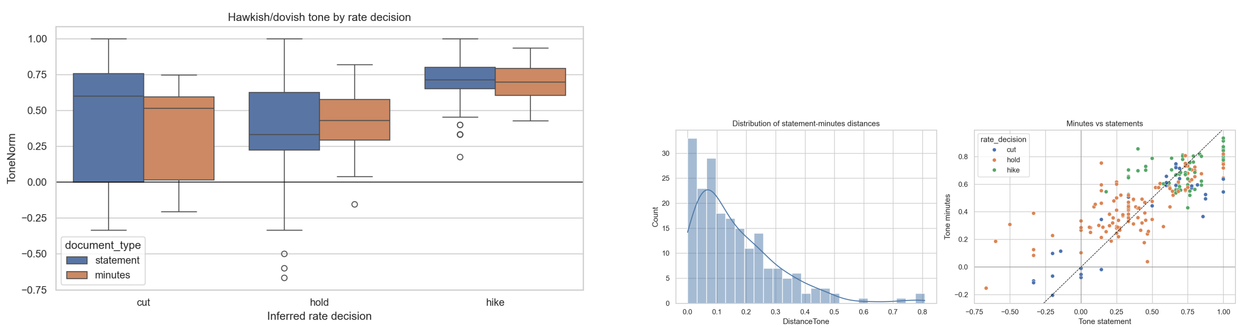


Figure 1: Normalized hawkish/dovish tone in FOMC statements and minutes.



(a) Tone by decision.

(b) Statement-minute distance.

Figure 2: Document tone by rate decision and meeting-level distance between statements and minutes.

5 Results

5.1 The Tone Signal Over Time

Figure 1 shows that the average tone is positive in both document types. The mean normalized score is 0.484 for statements and 0.507 for minutes. This does not mean that the Fed is always tightening. It means that, over the full sample, language related to inflation, price pressure and restrictive conditions is more frequent than language related to easing or weakness. The score falls in the early 2000s and in crisis periods, then rises during the 2015–2019 normalization and the post-Covid inflation episode.

The masked score is very close to the main score: the means are 0.474 for statements and 0.501 for minutes. The policy signal is therefore not driven only by direct decision sentences. The same gradient appears at sentence level: policy-relevant sentences account for about 31.0% of statement sentences and 29.7% of minute sentences, and their balance is highest during hike meetings.

The time series also illustrates why a single average score is not enough. During the zero lower bound period, policy was highly accommodative, yet the documents continued to discuss price stability and inflation expectations. During the post-Covid inflation period, the language becomes more directly hawkish because the inflation diagnosis and the tightening signal point in the same direction. The dictionary therefore behaves less like a generic sentiment index and more like a stance indicator: it reacts to the mixture of inflation-risk language and accommodation language, not to whether the economic outlook is described as “good” or “bad”.

Table 3: Average normalized tone by rate decision.

Document type	Cut	Hold	Hike
Statement	0.418	0.384	0.705
Minutes	0.368	0.442	0.700

(a) Tone and VIX over time.

(b) Tone by macro regime.

Figure 3: Tone across financial stress and macro-financial regimes.

5.2 Statements, Minutes and Rate Decisions

Hike meetings are clearly more hawkish than hold and cut meetings. Table 3 reports average scores. For statements, hike meetings have an average tone of 0.705, compared with 0.384 for holds and 0.418 for cuts. For minutes, the corresponding values are 0.700, 0.442 and 0.368. The lower average score for cut minutes is economically intuitive: minutes written around easing episodes emphasize weakness, downside risks and support more than hike minutes do.

The statement-minute distance is

$$\text{DistanceTone}_t = |\text{ToneNorm}_{minutes,t} - \text{ToneNorm}_{statement,t}|. \tag{3}$$

Its mean is 0.163 and its median is 0.122. A paired comparison does not indicate a large systematic difference in average tone between statements and minutes, but the meeting-level distribution is informative. Several meetings, especially in the early-2000s easing cycle, display large gaps. In those cases, the statement can be strongly dovish while the minutes remain more balanced because they record competing risks or a more detailed debate.

This distinction is conceptually important. Statements are written as coordinated public signals. They must summarize the Committee's decision in a compact and market-readable form. Minutes, by contrast, are designed to document the reasoning behind the decision. They can contain hawkish and dovish arguments in the same document, especially when the Committee faces a trade-off between inflation concerns and real-activity weakness. A large statement-minute distance should therefore not be interpreted automatically as inconsistency. It is better read as evidence that the later document adds nuance to the immediate signal.

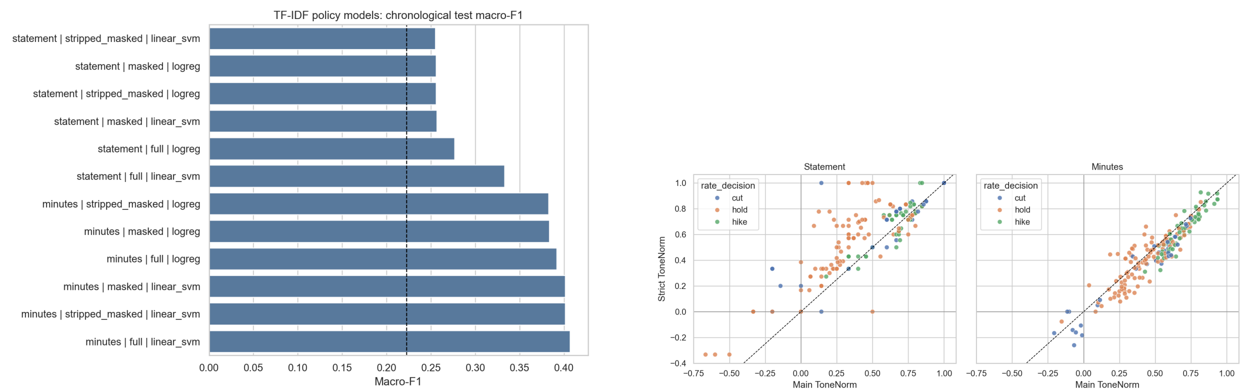
5.3 Financial and Political Context

High-volatility meetings are associated with lower tone scores. In meetings below the upper VIX quartile, the average score is 0.542 for statements and 0.549 for minutes. In high-VIX meetings, the averages fall to 0.311 and 0.382. This pattern is consistent with the Fed using more dovish or support-oriented language during episodes of financial stress. In contrast, election windows show little average difference: statement tone is 0.494 outside election windows and 0.459 inside them; minute tone is 0.511 outside and 0.497 inside.

The macro-regime analysis is consistent with the time series. The normalization period and the post-Covid inflation period are among the most hawkish regimes. The Covid period and the early-2000s easing cycle are less hawkish. The global financial crisis is not the most dovish regime in the dictionary score because those documents also contain frequent references to inflation, price stability and financial strains. This illustrates an important point: a text can describe severe weakness while still preserving an inflation-risk vocabulary.

Table 4: Descriptive OLS model for normalized tone. The reference category is a statement during a hold meeting.

Variable	Coefficient	p-value
Intercept	0.406	0.000
Minutes	0.023	0.376
Hike decision	0.255	0.000
Cut decision	0.012	0.755
VIX, standardized	-0.082	0.000
Election window	0.002	0.945



(a) TF-IDF chronological test macro-F1.

(b) Main and strict dictionary scores.

Figure 4: Representation and dictionary robustness checks.

The absence of a strong election-window pattern is also informative. In this descriptive design, the tone series does not suggest an obvious shift in the language of policy around presidential or midterm elections. This should not be overinterpreted as evidence that politics never affects communication. The election variable is deliberately simple, and the number of meetings inside election windows is limited. The result is narrower: conditional on this corpus and this dictionary score, the most visible variation is linked to policy and financial conditions, not to the election-window indicator.

5.4 Regression Evidence

Table 4 summarizes the descriptive associations in a pooled document panel. The hike coefficient is large and statistically significant, which confirms that the tone score is aligned with the most tightening-oriented decision class. The VIX coefficient is negative and significant. The minutes dummy is positive but not statistically significant, consistent with the paired comparison. The election coefficient is essentially zero in this specification.

The cut coefficient is small in the pooled specification, even though cut minutes are visibly less hawkish in the grouped averages. This is a useful reminder that rate cuts are heterogeneous. Some cuts occur in crisis periods dominated by support and downside-risk language; others occur when inflation concerns have not fully disappeared. The tone score therefore captures the communication surrounding the decision, not only the sign of the rate change.

5.5 NLP Model Checks

The TF-IDF models provide a benchmark for whether rate-decision information is present in the text beyond the handcrafted score. The best chronological test performance is obtained with minutes and a linear SVM: 0.550 accuracy and 0.407 macro-F1. The same model on masked minutes reaches the same accuracy and 0.401 macro-F1, which again suggests that the decision signal is not confined to the explicit announcement sentence. Statement models perform worse, with the best macro-F1 around 0.333. This gap is plausible because statements are short and formulaic, while minutes contain more evidence about the economic reasoning behind the decision.

The TF-IDF coefficients are useful for auditing the model. Hike-associated terms include inflation, prices, energy prices

Table 5: Summary of representation checks.

Check	Statement result	Minutes result
Masked mean tone	0.474	0.501
Strict dictionary correlation with main score	0.839	0.935
Best TF-IDF chronological macro-F1	0.333	0.407
LSA prototype correlation with dictionary tone	0.407	0.150
Sentence-BERT prototype correlation with dictionary tone	0.043	0.048

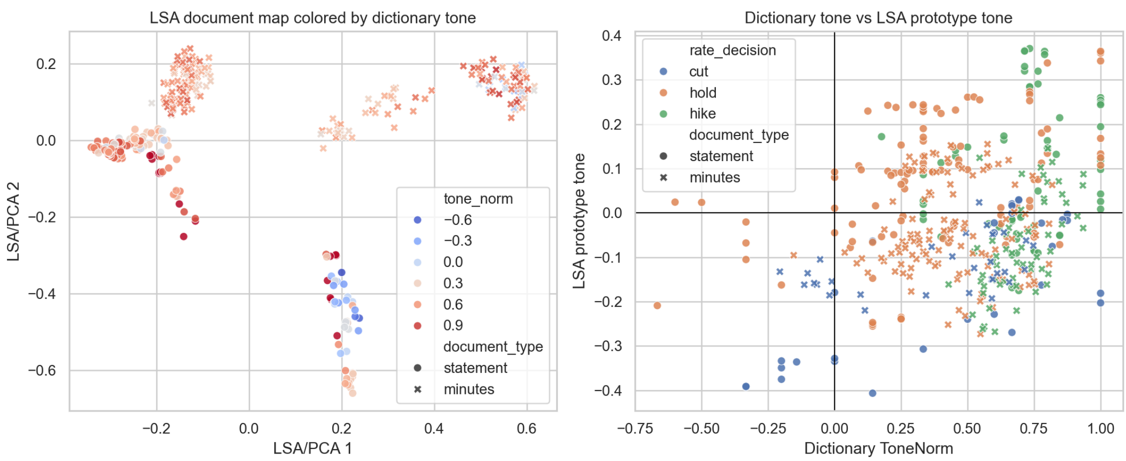


Figure 5: LSA document geometry and prototype-based tone.

and point increases. Cut-associated terms include discount-rate reductions and crisis-era action vocabulary. Hold-associated terms include purchases, recovery, accommodative stance and asset programs. The terms are economically interpretable, but they also reveal a limitation: a flexible text classifier can partly learn episode-specific vocabulary, not only policy stance.

Dictionary sensitivity is checked by removing broad words such as generic *increase*, *pressure* and *support*. The main and strict scores remain strongly related. The correlation is 0.839 for statements and 0.935 for minutes. The mean absolute difference is larger for statements (0.159) than for minutes (0.079), which is expected because short documents are more sensitive to small dictionary changes. The ordering by rate decision remains stable.

Table 5 summarizes the main robustness checks. The strict dictionary result supports the stability of the lexical score. The TF-IDF result shows that minutes contain more predictive information about the policy decision than statements do. The embedding correlations are weaker, especially for minutes, which suggests that dense semantic structure is not equivalent to the explicit hawkish/dovish concept measured by the dictionary.

5.6 Embedding Robustness

Figure 5 compares the dictionary score with an LSA prototype score. The first 30 LSA components explain 51.5% of TF-IDF variance. The correlation between LSA prototype tone and dictionary tone is 0.407 for statements and 0.150 for minutes. This positive but moderate relationship suggests that LSA captures some policy-tone information, especially in short statements, while minutes' dense geometry is shaped by broader topics and document structure.

The nearest-neighbor inspection in LSA space gives the same message. For selected meetings, neighboring documents often come from similar policy episodes, such as crisis meetings, zero lower bound communications, or inflation-tightening periods. This is useful for qualitative exploration: embeddings can retrieve comparable communication

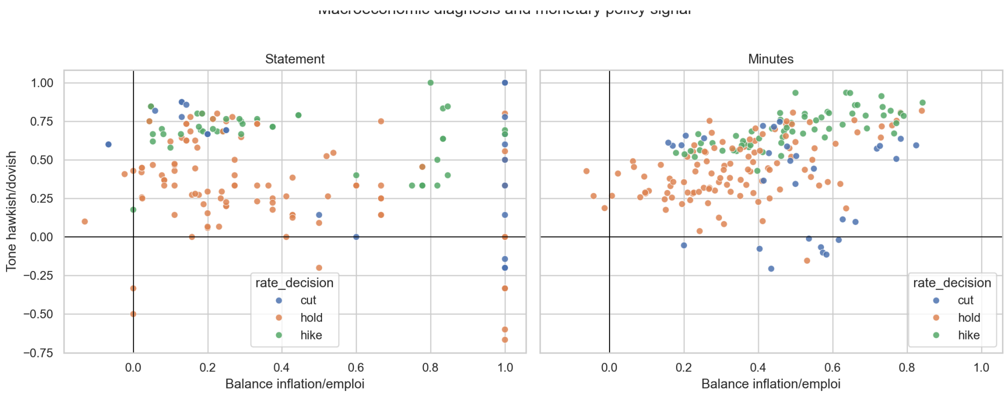


Figure 6: Inflation/employment balance and hawkish/dovish tone.

episodes without relying on the dictionary. However, the neighbors are not always ordered by hawkishness. They may instead be close because they share institutional vocabulary, document format or crisis-specific topics. This is precisely why the embedding analysis is treated as a robustness check rather than as the primary measure.

The Sentence-BERT prototype check produces very weak correlations with the dictionary tone: 0.043 for statements and 0.048 for minutes. This does not mean that Sentence-BERT is uninformative. It means that a generic contextual embedding, combined with short hawkish and dovish prototypes, does not reproduce the explicit monetary-policy lexicon. In this application, Sentence-BERT is better viewed as a tool for qualitative nearest-neighbor inspection or clustering of communication episodes than as a direct substitute for the interpretable dictionary score.

This result is also methodologically useful. Sentence-BERT embeddings are trained to encode broad sentence similarity, not to estimate central-bank stance. Without fine-tuning or labelled examples, the model has no reason to place *price stability*, *restrictive stance*, and *downside risks* on the exact monetary-policy axis needed here. The weak prototype correlations therefore support the choice of a domain-specific measure. A supervised embedding model could be a natural extension, but it would require expert labels at the sentence or paragraph level.

5.7 From Economic Diagnosis to Policy Signal

The final analysis relates the hawkish/dovish score to a separate inflation/employment balance. The correlation between mandate balance and tone is small for statements (-0.042) but positive for minutes (0.405). This difference is substantive. Statements are concise and strongly calibrated around the immediate policy message, so the diagnostic balance and the policy signal can diverge. Minutes provide more space for connecting inflation, labor-market conditions and the policy stance; the two dimensions therefore move together more clearly.

Most meetings fall in the inflation-hawkish quadrant. This reflects both the importance of price stability in FOMC communication and the design of the hawkish/dovish dictionary. More interesting are the off-diagonal cases: meetings where inflation language is salient but the tone is relatively dovish, or where employment concerns coexist with hawkish policy language. These cases are useful for qualitative interpretation because they identify moments when the diagnosis and the policy signal do not point in the same direction.

Economically, this two-dimensional view is useful because the Fed's dual mandate and the policy stance are related but not identical. A document can be inflation-focused because the Committee is explaining high realized inflation, but the policy signal can still be cautious if financial conditions are fragile. The opposite case can occur when labor-market language remains prominent while the Committee signals that policy must stay restrictive. The scatter plot therefore turns a one-dimensional tone series into a map of communication regimes.

6 Discussion

The main conclusion is that a domain-specific dictionary provides a transparent and economically interpretable measure of monetary-policy tone. It recovers the expected ordering of rate decisions, is robust to decision-sentence masking, and remains stable under a stricter dictionary. It also delivers interpretable meeting-level distances between statements and minutes.

The findings also speak to the role of document type. The statement is the high-salience message released immediately after a meeting. It compresses the decision and the Committee's assessment into a short, carefully negotiated text. The minutes, by contrast, are a delayed explanation of the discussion. The fact that their average tone is not statistically different from statements should not hide their analytical value. Their length makes them more useful for detecting the reasons behind the signal, and their meeting-level distance from the statement helps identify where the public message was later qualified by more detailed arguments.

The model is nevertheless limited. Dictionary scores depend on the chosen terms, and some words are context-dependent. A term such as *support* may describe policy accommodation, but it can also appear in a neutral institutional phrase. Conversely, a sentence can be hawkish without using any dictionary term. The strict-dictionary and sentence-level checks reduce this concern but do not remove it.

Another limitation concerns the unit of observation. The analysis treats each document as one observation, but minutes contain multiple speakers, topics and sections. A more granular model could separate staff reviews, participant views and policy discussion. It could also distinguish uncertainty language from stance language. This would matter in meetings where the Fed describes weak activity, high uncertainty and inflation risk at the same time.

The supervised and embedding checks clarify what the dictionary does and does not measure. TF-IDF classifiers confirm that policy-decision information is present in the text, especially in minutes, but they also learn episode-specific vocabulary. LSA and Sentence-BERT capture semantic neighborhoods and macroeconomic episodes, yet they do not naturally reproduce the hawkish/dovish scale without stronger supervision. A stronger extension would annotate sentences manually as hawkish, dovish or neutral and train a supervised classifier on expert labels. Another extension would test market reactions around meetings to distinguish descriptive tone from communication surprises.

The external validity of the approach is promising but not automatic. The same pipeline could be applied to European Central Bank press conferences or Bank of England minutes, but the dictionaries would need to be adapted to institutional language and policy frameworks. Even within the Fed corpus, vocabulary changes over time. Terms related to asset purchases, forward guidance and balance-sheet policy become central after 2008. A static dictionary is therefore easy to interpret, but future work could allow the vocabulary to evolve while preserving an auditable mapping to hawkish and dovish concepts.

7 Conclusion

This paper measures the hawkish/dovish signal in FOMC statements and minutes from 2000 to 2024. The empirical results show that hawkish language is highest around rate hikes, lower in high-VIX periods, and not visibly different around election windows. Minutes and statements have similar average tone but can differ substantially at the meeting level. Robustness checks based on decision masking, sentence tagging, TF-IDF models, strict dictionaries, LSA and Sentence-BERT support the central interpretation: the dictionary score is a useful, transparent measure of monetary-policy signal, while dense embeddings are better suited to studying broader semantic structure and communication regimes.

References

A. S. Blinder, M. Ehrmann, M. Fratzscher, J. De Haan, and D.-J. Jansen. Central bank communication and monetary policy: A survey of theory and evidence. *Journal of Economic Literature*, 46(4):910–945, 2008.

Board of Governors of the Federal Reserve System. FOMC calendars, statements and minutes. Online archive, 2024.

Cboe Global Markets. VIX historical price data. Online data file, 2026.

S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman. Indexing by latent semantic analysis. *Journal of the American Society for Information Science*, 41(6):391–407, 1990.

M. Gentzkow, B. T. Kelly, and M. Taddy. Text as data. *Journal of Economic Literature*, 57(3):535–574, 2019.

S. Hansen, M. McMahon, and A. Prat. Transparency and deliberation within the FOMC: A computational linguistics approach. *The Quarterly Journal of Economics*, 133 (2):801–870, 2018.

T. Loughran and B. McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65, 2011.

D. O. Lucca and F. Trebbi. Measuring central bank communication: An automated approach with application to FOMC statements. Working Paper 15367, National Bureau of Economic Research, 2009.

N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992, 2019.